Quantum Confinement in a Homogeneous $SU(3)$ Gauge–Scalar Model

Codex mathematical verification and continuation

4 September 2026 — research draft

Abstract

We canonically quantize the complete spatially homogeneous mechanics of an $SU(3)$ gauge field coupled to a traceless Hermitian adjoint scalar Φ and a complex triplet b, retaining all thirty-eight real configuration coordinates. Its potential contains the original cubic $\det\Phi - b^*\Phi b$ and quartic term in $s = \operatorname{tr}\Phi^2 + 2b^*b$; no minimum-manifold constraint is imposed. An explicit root-frequency calculation in the metric $-\operatorname{tr}$ gives a coercive quadratic-form bound despite the commuting classical valleys. Together with a sharp cubic estimate, this proves compact resolvent of the full coupled Hamiltonian. We derive the Gauss generators and their domains, construct the constant-$SU(3)$ invariant Hilbert space, and prove that its ground state is unique and its first excitation gap is strictly positive for every fixed positive spatial volume, gauge coupling, quartic coupling and Planck constant. No numerical or volume-uniform gap is claimed. The mechanism of bosonic matrix discreteness is classical; the purpose is its explicit application to this retained coupled model. We also prove the precise classical embedding into the torus field theory and exhibit a quartic Fourier-mode coupling that prevents a fixed discarded-mode vacuum from defining an automatic quantum subsector. Thus the finite-model result is not a continuum Yang–Mills mass-gap theorem. For the exact interacting two-Fourier-coordinate truncation, we then construct the actual positive quartic fibre ground state Ω_q. Its isometric embedding has zero Berry connection but a nonzero Born–Huang potential $\hbar^2\|\partial_q\Omega_q\|^2/(2\mathcal{V}_3)$ away from $q = 0$. We retain its full off-diagonal residual and prove an explicit finite-energy, physical-time comparison of order $\mathcal{V}_3^{-1/2}$ at fixed $\hbar, \Gamma, L_1, m^2$. The sufficient volume bound and every error constant are given; neither the mixed quartic interaction nor the pure discarded-mode quartic is removed. This correspondence is not a reducing subspace or a comparison with an unconstructed continuum Hamiltonian.

1 Quantum confinement in the homogeneous gauge–scalar model

The object quantized here is a precisely specified interacting matrix mechanics: three anti-Hermitian traceless gauge matrices, one Hermitian traceless adjoint scalar, and one complex triplet. It is obtained by spatially constant substitution in the gauge–scalar action used in the quaternionic continuation. All of these fields remain variables. In particular the triplet is not constrained to a four-real-dimensional slice or to a scalar minimum curve.

The conclusions are a compact-resolvent Hamiltonian, a unique spatially constant $SU(3)$-invariant ground state, and a strictly positive first excitation gap at fixed positive volume and couplings. The explicit form inequality (9) is the quantitative step. It combines the exact $SU(3)$ root frequencies with the full quartic scalar potential. Bosonic commutator-matrix discreteness is a classical result; the primary attribution and the complete proof in the retained metric are given below. The present use is to answer a concrete quantization question for the already specified adjoint-plus-triplet action, not to claim a new general discreteness mechanism.

For the classical comparison the spacetime signature is $(+---)$ and the original action density is

$$\mathcal{L} = \frac{1}{2g^2}\,\mathrm{tr}\,F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\,\mathrm{tr}(D_\mu\Phi D^\mu\Phi) + (D_\mu b)^* D^\mu b - V(\Phi, b),$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \quad D_\mu\Phi = \partial_\mu\Phi + [A_\mu, \Phi], \quad D_\mu b = \partial_\mu b + A_\mu b.$$

The potential is written in full below. The finite-model theorem does not require the construction of a compact complex structure on S^6, and it neither verifies nor refutes that separate global construction. The precise classical map to the torus field theory, its noninjective map on gauge orbits, and an explicit obstruction to one proposed quantum-subsector map are proved at the end.

1.1 The retained configuration space and action

Fix a flat spatial torus of volume $\mathcal{V}_3 > 0$, a gauge coupling $g^2 > 0$, and $\hbar > 0$. Retain $m^2 \geq 0$, $\Gamma > 0$ and $\kappa \in \mathbb{R}$. We quantize the finite-dimensional homogeneous mechanics, not a claimed invariant quantum sector of the full field theory. The configuration space, with no quotient already performed, is

$$\mathcal{Q} = \mathfrak{su}(3)^3 \oplus \mathrm{Herm}_3(\mathbb{C})_0 \oplus \mathbb{C}^3, \qquad \dim_{\mathbb{R}} \mathcal{Q} = 24 + 8 + 6 = 38.$$

Its elements are $q = (A_1, A_2, A_3, \Phi, b)$. On $\mathfrak{su}(3)$ use $(X, Y)_{\mathfrak{g}} = -\,\mathrm{tr}(XY)$; the scalar metric is exactly $\mathrm{tr}(d\Phi^2) + 2\,db^* db$. The trace-free condition on Φ is part of the linear state space.

For a homogeneous temporal connection A_0, set

$$E_i = \dot{A}_i + [A_0, A_i], \quad X = \dot{\Phi} + [A_0, \Phi], \quad v = \dot{b} + A_0 b, \quad s = \mathrm{tr}\,\Phi^2 + 2b^* b.$$

The spatial integral of the retained $(+---)$, anti-Hermitian density is

$$\begin{aligned} L_{\mathrm{hom}} &= \mathcal{V}_3 \left[\frac{1}{g^2} \sum_i \|E_i\|_{\mathfrak{g}}^2 + \frac{1}{2}\,\mathrm{tr}\,X^2 + v^* v - U(q) \right], \\ U(q) &= \frac{1}{g^2} \sum_{i<j} \|[A_i, A_j]\|_{\mathfrak{g}}^2 + \frac{1}{2} \sum_i \mathrm{tr}[A_i, \Phi]^2 + \sum_i \|A_i b\|^2 + V(\Phi, b), \\ V(\Phi, b) &= \frac{m^2}{2} s + \frac{\Gamma}{4} s^2 - \kappa(\det\Phi - b^*\Phi b). \end{aligned} \tag{1}$$

Here $[A_i, \Phi]$ is Hermitian, so its displayed squared trace is nonnegative. There is no velocity for A_0.

Proposition 1.1 (Canonical momenta, Hamiltonian and exact Gauss map)**.** *Use the canonical one-form*

$$\vartheta = \sum_i -\,\mathrm{tr}(\Pi_i\, dA_i) + \mathrm{tr}(\Pi_\Phi\, d\Phi) + 2\,\mathrm{Re}(\pi_b^* db).$$

The momenta and Hamiltonian are

$$\begin{aligned} \Pi_i &= \frac{2\mathcal{V}_3}{g^2} E_i, \qquad \Pi_\Phi = \mathcal{V}_3 X, \qquad \pi_b = \mathcal{V}_3 v, \\ H_{A_0} &= H_0 - \mathcal{M}_{A_0}, \\ H_0 &= \frac{g^2}{4\mathcal{V}_3} \sum_i \|\Pi_i\|_{\mathfrak{g}}^2 + \frac{1}{2\mathcal{V}_3}\,\mathrm{tr}\,\Pi_\Phi^2 + \frac{1}{\mathcal{V}_3}\pi_b^*\pi_b + \mathcal{V}_3 U(q). \end{aligned} \tag{2}$$

For $\xi \in \mathfrak{su}(3)$, write

$$K_\xi(q) = ([\xi, A_i]_{i=1}^3, [\xi, \Phi], \xi b), \qquad \mathcal{M}_\xi = \vartheta(K_\xi) = (\mathcal{G}, \xi)_{\mathfrak{g}}.$$

The exact matrix-valued momentum map is

$$\mathcal{G} = \sum_i [A_i, \Pi_i] - [\Phi, \Pi_\Phi] - (b\pi_b^* - \pi_b b^*)_{\mathrm{tf}} = \mathcal{V}_3 \left(\frac{2}{g^2} \sum_i [A_i, E_i] - (J_0)_{\mathrm{tf}} \right), \tag{3}$$

where $J_0 = [\Phi, X] + bv^* - vb^*$. Thus $\mathcal{G} = 0$ is precisely the full homogeneous Gauss equation.

Proof. Differentiate (1) using the stated duality pairings. In particular, differentiation of v^*v gives $2\mathcal{V}_3 \operatorname{Re}(v^* \delta \dot{b})$, explaining $\pi_b = \mathcal{V}_3 v$ rather than $2\mathcal{V}_3 v$ in this complex duality convention. The velocity is the covariant velocity minus K_{A_0}; its Legendre transform is consequently $H_0 - \vartheta(K_{A_0})$. The ordinary quadratic Legendre transform gives every coefficient of (2). Cyclicity gives

$$-\operatorname{tr}(\Pi_i[\xi, A_i]) = -\operatorname{tr}([A_i, \Pi_i]\xi), \quad \operatorname{tr}(\Pi_\Phi[\xi, \Phi]) = \operatorname{tr}([\Phi, \Pi_\Phi]\xi),$$

while anti-Hermiticity gives

$$2\operatorname{Re}(\pi_b^* \xi b) = \operatorname{tr}((b\pi_b^* - \pi_b b^*)\xi).$$

These identities prove the first expression for $\mathcal{G}$; substitution of the momenta proves the second. Variation of the independent A_0 multiplier imposes $\mathcal{G} = 0$, even after a temporal gauge is selected for the evolution. The primary momentum conjugate to A_0 is zero; A_0 is not an additional coordinate of $\mathcal{Q}$. Since $\mathcal{M}_\xi = p \cdot K_\xi$, canonical Poisson brackets give $\{\mathcal{M}_\xi, \mathcal{M}_\eta\} = \mathcal{M}_{[\xi,\eta]}$. Gauge invariance of every trace and norm in H_0 also gives $\{H_0, \mathcal{M}_\xi\} = 0$. □

Coordinates and measure without changing the metrics. Choose any fixed orthonormal basis T_a, $1 \le a \le 8$, for $-\operatorname{tr}$ on $\mathfrak{su}(3)$, and put $S_a = iT_a$. Then $\operatorname{tr}(S_a S_b) = \delta_{ab}$. Write

$$A_i = \sum_a a_{ia} T_a, \quad \Phi = \sum_a \phi_a S_a, \quad b_r = x_r + iy_r.$$

These are coordinates on the original spaces, not a rescaling of their metrics. The mass metric read from (1) is

$$M = \operatorname{diag}\left(\frac{2\mathcal{V}_3}{g^2} I_{24},\ \mathcal{V}_3 I_8,\ 2\mathcal{V}_3 I_6 \right).$$

We use its constant volume measure

$$d\mu_M = C_M\, d^{24}a\, d^8\phi\, d^3x\, d^3y, \qquad C_M = \sqrt{\det M} = \frac{2^{15}\mathcal{V}_3^{19}}{(g^2)^{12}},$$

retaining this constant rather than setting it to one. The auxiliary Hilbert space is $\mathscr{H} = L^2(\mathcal{Q}, d\mu_M; \mathbb{C})$. In these coordinates canonical momentum is $-i\hbar\partial$. Thus the exact kinetic operator is

$$T = -\frac{\hbar^2 g^2}{4\mathcal{V}_3}\Delta_a - \frac{\hbar^2}{2\mathcal{V}_3}\Delta_\phi - \frac{\hbar^2}{4\mathcal{V}_3}(\Delta_x + \Delta_y). \tag{4}$$

Equivalently the complex momentum is $\widehat{\pi}_{b_r} = -i\hbar\partial_{\bar{b}_r} = -i\hbar(\partial_{x_r} + i\partial_{y_r})/2$; its adjoint product is $-\hbar^2(\partial_{x_r}^2 + \partial_{y_r}^2)/4$. The Schrödinger equation is $i\hbar\partial_t\psi = H\psi$.

1.2 The retained cubic and a global scalar bound

Lemma 1.2 (Sharp cubic bound in the original complex coordinates)**.** *For every traceless Hermitian* $\Phi \in \mathrm{Herm}_3(\mathbb{C})$ *and* $b \in \mathbb{C}^3$*, put*

$$s = \operatorname{tr} \Phi^2 + 2b^* b, \qquad N = \det \Phi - b^* \Phi b.$$

Then

$$|N| \le \frac{s^{3/2}}{3\sqrt{6}}. \tag{5}$$

Consequently, with all original coefficients retained,

$$V(\Phi, b) = \frac{m^2 s}{2} + \frac{\Gamma s^2}{4} - \kappa N \;\ge\; \frac{\Gamma s^2}{8} - C_\kappa, \qquad C_\kappa = \frac{\kappa^4}{54\Gamma^3}, \tag{6}$$

whenever $m^2 \ge 0$*,* $\Gamma > 0$*, and* $\kappa \in \mathbb{R}$*. This is an inequality for the full potential, not its replacement by the right-hand side.*

Proof. Choose a unitary diagonalization of Φ and apply the same unitary to b. The inverse conjugation restores both original fields; it preserves s, N and introduces no restriction on either field. Write the three real eigenvalues as $\lambda_1, \lambda_2, \lambda_3$, with sum zero, and let $z_i = |b_i|^2$, $u = \sum_i z_i$. If $u > 0$, the original cubic is the convex combination

$$N = \sum_{i=1}^{3} \frac{z_i}{u} (\lambda_1 \lambda_2 \lambda_3 - u\lambda_i).$$

For one index i and the other two indices j, k, put $d_i = \lambda_j - \lambda_k$. Trace zero gives

$$\lambda_j \lambda_k = (\lambda_i^2 - d_i^2)/4, \qquad s = (3\lambda_i^2 + d_i^2)/2 + 2u.$$

Hence each term in parentheses is exactly

$$\lambda_i(\lambda_j \lambda_k - u) = \lambda_i^3 - \frac{s\lambda_i}{2}, \qquad |\lambda_i| \le \sqrt{2s/3}.$$

If $u = 0$, the same identity equals N for each i. If $s = 0$, both fields vanish and the assertion is immediate. For $s > 0$, the critical points of $x^3 - sx/2$ on $[-\sqrt{2s/3}, \sqrt{2s/3}]$ are $x = \pm\sqrt{s/6}$. At these two points and the two endpoints, its absolute value is $s^{3/2}/(3\sqrt{6})$. The elementary extreme-value theorem proves the bound for every term and hence for their convex combination. Equality is attained, for example, at $\Phi = \mathrm{diag}(2r, -r, -r)$, $b = 0$, $r \ne 0$, since then $s = 6r^2$ and $N = 2r^3$.

Put $\alpha = |\kappa|/(3\sqrt{6})$ and $t = \sqrt{s} \ge 0$. Keep $m^2 s/2$ as a nonnegative term when bounding the original potential. The remaining estimate is

$$V \ge \frac{\Gamma t^4}{8} + \left(\frac{\Gamma t^4}{8} - \alpha t^3 \right).$$

For $\alpha > 0$ the expression in parentheses has its global minimum at $t = 6\alpha/\Gamma$, with value $-54\alpha^4/\Gamma^3 = -\kappa^4/(54\Gamma^3)$. For $\alpha = 0$ it is nonnegative. This proves (6) including both signs of κ. □

Theorem 1.3 (Friedrichs realization with an explicit domain)**.** *Let* $W(q) = \mathcal{V}_3 U(q)$ *and*

$$D = \mathrm{diag}\left(\frac{\hbar^2 g^2}{4\mathcal{V}_3} I_{24},\ \frac{\hbar^2}{2\mathcal{V}_3} I_8,\ \frac{\hbar^2}{4\mathcal{V}_3} I_6 \right), \qquad C = \frac{\mathcal{V}_3 \kappa^4}{54\Gamma^3}.$$

The differential expression $T + W = -\nabla \cdot D\nabla + W$ on $C_c^\infty(\mathcal{Q})$ has the following well-defined semibounded Friedrichs realization H. Its closed form and form domain are

$$\begin{aligned} \mathfrak{h}[\psi] &= \int \Big((\nabla\psi)^* D \nabla\psi + W|\psi|^2 \Big)\, d\mu_M, \\ \mathscr{D} &= H^1(\mathcal{Q}, d\mu_M) \cap L^2(\mathcal{Q}, (W+C)d\mu_M), \\ D(H) &= \{u \in \mathscr{D} : -\nabla \cdot D\nabla u + Wu \in L^2(\mathcal{Q}, d\mu_M) \text{ in distributions}\}. \end{aligned} \tag{7}$$

The operator is self-adjoint, bounded below by $-C$, and C_c^∞ is a form core. No compactness assertion is needed for this construction.

Proof. Lemma 1.2 proves directly in the original complex coordinates that $V \geq \Gamma s^2/8 - \kappa^4/(54\Gamma^3)$. All the other terms of U are nonnegative. In particular $W+C$ is a continuous nonnegative polynomial, locally bounded everywhere.

The norm

$$\|u\|_{\mathscr{D}}^2 = \int \Big((\nabla u)^* D \nabla u + (W + C + 1)|u|^2 \Big)\, d\mu_M$$

is complete. Indeed a Cauchy sequence converges in H^1 and in the weighted L^2 space, and their limits agree on every compact set. This proves closedness. Smooth radial cutoffs converging to one approximate any $u \in \mathscr{D}$ in this norm: the potential and gradient tails tend to zero, and the cutoff-gradient term is bounded by a constant times $R^{-2}\|u\|^2$. On a fixed compact set ordinary mollification converges in H^1, and the bounded weight there gives weighted convergence. Thus C_c^∞ is a form core and the form is densely defined.

For clarity one may construct its self-adjoint operator directly. The positive form $\mathfrak{h} + (C+1)\langle\cdot,\cdot\rangle$ is an inner product on $\mathscr{D}$. Riesz representation assigns to every $f \in \mathscr{H}$ a unique $u = Rf \in \mathscr{D}$ such that

$$\mathfrak{h}(u, v) + (C+1)\langle u, v\rangle = \langle f, v\rangle \quad (v \in \mathscr{D}).$$

It follows that R, viewed on $\mathscr{H}$, is bounded, self-adjoint, positive and injective; its range is dense because its kernel is zero. The spectral theorem for bounded self-adjoint operators gives the self-adjoint inverse R^{-1} on $\operatorname{Ran} R$. Set $H = R^{-1} - (C+1)$. This is precisely the Friedrichs operator of the original form core. Integration by parts on compact test functions identifies its distributional expression. Conversely, for $u \in \mathscr{D}$ whose displayed distribution is $f \in L^2$, that integration-by-parts identity extends from the form core to all of $\mathscr{D}$; the Riesz characterization gives $u \in D(H)$, $Hu = f$. This proves exactly the asserted domain. It does not replace the domain by an unjustified separate requirement $Wu, \Delta u \in L^2$. □

Theorem 1.4 (Spatially constant Gauss constraints and averaging)**.** *For $g \in SU(3)$, let*

$$R_g q = (gA_i g^{-1}, g\Phi g^*, gb), \qquad (\mathcal{U}(g)\psi)(q) = \psi(R_{g^{-1}} q).$$

This is a strongly continuous unitary representation on $\mathscr{H}$. For each $\xi \in \mathfrak{su}(3)$, the operator

$$\widehat{\mathcal{M}}_\xi = -i\hbar K_\xi \cdot \nabla \quad \text{on } C_c^\infty(\mathcal{Q})$$

is essentially self-adjoint. Its closure has domain

$$D(\widehat{\mathcal{M}}_\xi) = \{\psi \in \mathscr{H} : K_\xi \cdot \nabla\psi \in \mathscr{H} \text{ distributionally}\}, \qquad \mathcal{U}(e^{t\xi}) = e^{-it\widehat{\mathcal{M}}_\xi/\hbar}.$$

The spatially constant physical Hilbert space and projection are

$$\mathscr{H}_{\text{phys}} = \{\psi : \mathcal{U}(g)\psi = \psi \ \forall g \in SU(3)\} = \bigcap_{a=1}^{8} \ker \widehat{\mathcal{M}}_{T_a},$$

$$\mathsf{P}\psi = \frac{1}{\mathcal{V}_\mu}\int_{SU(3)} \mathcal{U}(g)\psi \, d\mu(g), \qquad \mathcal{V}_\mu = \int_{SU(3)} d\mu(g) > 0. \tag{8}$$

Here the Haar mass is retained explicitly. The space is infinite dimensional. The set $C_c^\infty(\mathcal{Q})^{SU(3)} = \mathsf{P}C_c^\infty(\mathcal{Q})$ *is dense in it and is a form core for its physical Hamiltonian*

$$H_{\mathrm{phys}} = H|_{\mathscr{H}_{\mathrm{phys}}}, \qquad D(H_{\mathrm{phys}}) = D(H) \cap \mathscr{H}_{\mathrm{phys}}.$$

Thus the physical Hamiltonian is self-adjoint and is exactly the Friedrichs operator of the restricted form; no quotient-coordinate measure or singular-orbit boundary condition has been inserted.

Proof. Each R_g preserves the three original inner products, hence the mass metric, its measure and the kinetic form. It preserves U by conjugation invariance of the trace/determinant and unitarity of the triplet action. Continuity holds first on compact smooth functions and then on L^2 by density and unitarity. The vector field K_ξ is linear with zero divergence: its flow is an orthogonal linear map for each block. Its action on compact smooth functions is therefore skew-adjoint under integration, so $-i\hbar K_\xi \cdot \nabla$ is symmetric. Differentiating the displayed unitary action gives the stated generator sign. Stone's theorem gives a self-adjoint generator. Its domain is the displayed distributional domain: one direction follows by differentiation against test functions; for the converse, distributional integration along the flow gives

$$\mathcal{U}(e^{t\xi})\psi - \psi = -\int_0^t \mathcal{U}(e^{s\xi})(K_\xi \cdot \nabla\psi)\, ds$$

as an L^2 identity, and strong continuity gives differentiability.

To verify that the original compact smooth domain is a core, use smooth cutoffs depending on the blockwise invariant Euclidean radius. They commute with K_ξ. Convolution with a radial smooth kernel also commutes with its orthogonal flow. Cutoff followed by convolution therefore approximates both ψ and $K_\xi \cdot \nabla\psi$ in L^2. This proves graph-core density and essential self-adjointness. On that common smooth core,

$$[K_\xi \cdot \nabla, K_\eta \cdot \nabla] = -K_{[\xi,\eta]} \cdot \nabla, \qquad [\widehat{\mathcal{M}}_\xi, \widehat{\mathcal{M}}_\eta] = i\hbar\widehat{\mathcal{M}}_{[\xi,\eta]}.$$

There is no divergence correction or ordering anomaly.

An invariant vector lies in every generator domain with derivative zero. Conversely the eight vanishing distributional derivatives give the vanishing derivative for every ξ, hence invariance under every one-parameter subgroup. These generate connected $SU(3)$. For example each unitary determinant-one matrix is unitarily diagonalizable with eigenphases whose sum can be chosen to be zero, so it is the exponential of a trace-free anti-Hermitian matrix. This proves the kernel characterization with its domain included.

The Bochner integral in (8) is bounded in norm by one. Haar invariance and inversion imply $\mathsf{P}^2 = \mathsf{P} = \mathsf{P}^*$, with range precisely the invariant vectors. Averaging a compact smooth function remains smooth and has compact support, because the union of its supports under a compact group is compact. Applying P to any compact-smooth approximation proves L^2 density. Nonzero smooth functions supported in pairwise disjoint annuli of the invariant squared radius are invariant and orthogonal. This proves infinite dimensionality by explicit test states.

The form and its domain are $\mathcal{U}(g)$-invariant. The Riesz characterization above therefore gives $\mathcal{U}(g)R = R\mathcal{U}(g)$; averaging gives $\mathsf{P}R = R\mathsf{P}$. Functional calculus proves that P reduces H, so its restriction has the claimed self-adjoint domain. Jensen's inequality for the positive form $\mathfrak{h}+(C+1)\|\cdot\|^2$ makes P a contraction in the form norm. Averaging a form-core approximation of an invariant vector proves the final form-core statement. □

An explicit quantum bound for the three retained gauge matrices

Let $\mathfrak{g} = \mathfrak{su}(3)$ as a real eight-dimensional vector space, with the retained metric

$$\langle X, Y\rangle = -\operatorname{tr}(XY), \qquad \|X\|^2 = -\operatorname{tr}(X^2).$$

For anti-Hermitian X, Y, the trace of XY is real, by taking its conjugate and using cyclicity. The metric is positive definite because $-\operatorname{tr}(X^2) = \operatorname{tr}(X^*X)$. Lebesgue measure and all Laplacians below are those of this metric, not of the negative Killing form. Write $A = (A_1, A_2, A_3) \in \mathfrak{g}^3$. For two arbitrary retained numbers $a_g, b_g > 0$, consider on $L^2(\mathfrak{g}^3, dA)$ the nonnegative quadratic form

$$q_{a_g,b_g}[\psi] = a_g \sum_{i=1}^{3} \int \|\nabla_{A_i}\psi\|^2 \, dA + b_g \sum_{1\le i<j\le 3} \int \|[A_i, A_j]\|^2 |\psi|^2 \, dA, \tag{MQ.1}$$

initially on $C_c^\infty(\mathfrak{g}^3; \mathbb{C})$. The coefficients a_g, b_g are independent retained positive numbers; their subscripts distinguish them from the scalar data. No assertion about the scalar coupling or a Gauss-reduced operator is included in the theorem below.

Lemma 1.5 (The exact adjoint frequency sum, including root walls)**.** *Let $X \in \mathfrak{g}$, and let $\theta_1 \le \theta_2 \le \theta_3$ be the real eigenvalues of $-iX$, with $\sum_p \theta_p = 0$. The positive semidefinite real operator $K_X = -\operatorname{ad}_X^2$ has its eight eigenvalues, with real multiplicity, equal to*

$$0, 0, \quad (\theta_1-\theta_2)^2, (\theta_1-\theta_2)^2, \quad (\theta_1-\theta_3)^2, (\theta_1-\theta_3)^2, \quad (\theta_2-\theta_3)^2, (\theta_2-\theta_3)^2. \tag{MQ.2}$$

In particular, zeros in this list remain zeros at repeated eigenvalues. Define the frequency sum with real multiplicities by $\mathcal{S}(X) = \operatorname{tr}_{\mathbb{R}\mathfrak{g}} \sqrt{K_X}$. Then

$$\mathcal{S}(X) = 2\sum_{p<q} |\theta_p - \theta_q| = 4(\theta_3 - \theta_1), \qquad 2\sqrt{6}\,\|X\| \le \mathcal{S}(X) \le 4\sqrt{2}\,\|X\|. \tag{MQ.3}$$

Both constants are sharp. The lower equality at nonzero X occurs exactly when two eigenvalues coincide; the upper equality occurs exactly when the eigenvalues are $(-r, 0, r)$ for some $r \ge 0$. If each positive root is counted only once instead of using real multiplicity, the sum is $\mathcal{S}(X)/2$ and its sharp lower constant is $\sqrt{6}$.

Proof. Cyclicity gives $\langle [X,Y], Z\rangle = -\langle Y, [X,Z]\rangle$. Thus ad_X is skew-adjoint and $\langle Y, K_X Y\rangle = \|[X,Y]\|^2$. Unitary diagonalization of X is an isometry for the retained metric. Multiplication of that unitary by a scalar phase adjusts its determinant to one without changing its conjugation, so this diagonalization can also be performed by $SU(3)$.

For diagonal $X = i\operatorname{diag}(\theta_1, \theta_2, \theta_3)$, use the orthonormal Cartan vectors

$$D_1 = \frac{i}{\sqrt{2}}\operatorname{diag}(1, -1, 0), \qquad D_2 = \frac{i}{\sqrt{6}}\operatorname{diag}(1, 1, -2),$$

and, for $p < q$, the orthonormal pair

$$U_{pq} = \frac{E_{pq} - E_{qp}}{\sqrt{2}}, \qquad V_{pq} = \frac{i(E_{pq} + E_{qp})}{\sqrt{2}}.$$

Direct multiplication gives $[X, U_{pq}] = (\theta_p - \theta_q)V_{pq}$ and $[X, V_{pq}] = -(\theta_p - \theta_q)U_{pq}$, while D_1, D_2 commute with X. These eight orthonormal vectors span $\mathfrak{g}$, proving (MQ.2) and the exact sum in (MQ.3).

Put $r=\theta_2-\theta_1\geq 0$ and $s=\theta_3-\theta_2\geq 0$. The original eigenvalues are exactly

$$\theta_1=-(2r+s)/3,\quad \theta_2=(r-s)/3,\quad \theta_3=(r+2s)/3,\quad \|X\|^2=\frac{2}{3}(r^2+rs+s^2),\quad \mathcal{S}(X)=4(r+s).$$

Consequently

$$\mathcal{S}(X)^2-24\|X\|^2=16rs,\qquad 32\|X\|^2-\mathcal{S}(X)^2=\frac{16}{3}(r-s)^2.$$

Their nonnegativity proves the bounds and their vanishing proves every equality case. In particular the lower bound does not fail at a repeated eigenvalue: at a nonzero wall exactly four real frequencies remain positive. At $X=0$ both sides vanish. □

Lemma 1.6 (Slice oscillators with all coefficients)**.** *For fixed $X\in\mathfrak{g}$ and $\alpha,\beta>0$,*

$$-\alpha\Delta_Y+\beta\|[X,Y]\|^2\ \geq\ \sqrt{\alpha\beta}\,\mathcal{S}(X)\quad \text{on } L^2(\mathfrak{g},dY) \tag{MQ.4}$$

as a quadratic-form inequality. No positive frequency is assigned to the commuting coordinates.

Proof. For $\omega\geq 0$ and $f\in C_c^\infty(\mathbb{R};\mathbb{C})$, integration by parts proves the exact identity

$$\int|\sqrt{\alpha}f'+\sqrt{\beta}\,\omega y f|^2\,dy=\alpha\int|f'|^2\,dy+\beta\omega^2\int y^2|f|^2\,dy-\sqrt{\alpha\beta}\,\omega\int|f|^2\,dy.$$

Indeed $2\,\mathrm{Re}(f'\bar f)=(|f|^2)'$ and the boundary term vanishes. Thus each real oscillator contributes $\sqrt{\alpha\beta}\,\omega$, including the correct zero contribution when $\omega=0$.

Diagonalize the real symmetric K_X in an orthonormal basis. This orthogonal coordinate change preserves measure and Δ_Y, and the potential becomes the sum $\sum_{\nu=1}^{8}\omega_\nu^2y_\nu^2$ with the frequencies in (MQ.2). Apply the preceding identity in each coordinate, integrate the other coordinates, and sum to obtain (MQ.4). This proof is for each fixed X. It requires no differentiable choice of the diagonalizing basis as X varies, and introduces no derivatives in X. □

Theorem 1.7 (An explicit coercive form bound and compact resolvent)**.** *The closure of (MQ.1) has domain*

$$\mathcal{D}(q_{a_g,b_g})=\left\{\psi\in H^1(\mathfrak{g}^3):\sum_{i<j}\int\|[A_i,A_j]\|^2|\psi|^2\,dA<\infty\right\}. \tag{MQ.5}$$

Let H_A be its nonnegative Friedrichs operator. Its formal expression on compactly supported smooth functions is exactly

$$H_A=-a_g\sum_i\Delta_{A_i}+b_g\sum_{i<j}\|[A_i,A_j]\|^2. \tag{MQ.6}$$

For every $0\leq\vartheta<1$ the following form bound holds:

$$q_{a_g,b_g}[\psi]\geq\vartheta a_g\int\|\nabla\psi\|^2\,dA+2\sqrt{6(1-\vartheta)a_gb_g}\int\left(\sum_i\|A_i\|\right)|\psi|^2\,dA. \tag{MQ.7}$$

In particular, retaining half of the original kinetic energy gives

$$H_A\geq-\frac{a_g}{2}\Delta_{\mathfrak{g}^3}+2\sqrt{3a_gb_g}\sum_i\|A_i\|. \tag{MQ.8}$$

The form-domain inclusion into $L^2(\mathfrak{g}^3)$ is compact. Consequently $(H_A+\mu I)^{-1}$ is compact for every $\mu>0$, and H_A has only isolated eigenvalues of finite multiplicity, with no finite accumulation point.

Proof. The potential in (MQ.6) is a continuous nonnegative polynomial. The right side of (MQ.5), equipped with $\|\psi\|_2^2 + q_{a_g,b_g}[\psi]$, is complete: an associated Cauchy sequence converges in H^1 and also after multiplication by the square root of the potential; an almost-everywhere convergent subsequence identifies the second limit as that product. It is densely defined since it contains C_c^∞. Smooth cutoffs equal to one on a ball of radius R, zero outside the ball of radius $2R$, with derivative bounded by a constant times R^{-1}, approximate each member in this norm as $R \to \infty$. The derivative error tends to zero by the H^1 norm and the R^{-1} bound; the weighted error tends to zero by integrability. For each cutoff, convolution with a compactly supported smooth approximate identity converges in H^1, and also in weighted L^2 because the potential is bounded on the common compact support. This proves the stated core and closure.

For the lower bound start on this core and split the quadratic form exactly as

$$q_{a_g,b_g}[\psi] = \vartheta a_g \|\nabla\psi\|_2^2 + \sum_{\substack{i,j\in\{1,2,3\}\\ i\neq j}} \left\{ \frac{(1-\vartheta)a_g}{2} \|\nabla_{A_j}\psi\|_2^2 + \frac{b_g}{2}\int \|[A_i,A_j]\|^2 |\psi|^2 \, dA \right\}. \tag{MQ.9}$$

Every derivative appears in two directed pairs, and every unordered potential pair also appears twice. Hence no kinetic or potential term has been counted more than its original coefficient. Apply (MQ.4) inside each directed summand with $X = A_i$, $Y = A_j$, $\alpha = (1-\vartheta)a_g/2$, $\beta = b_g/2$. Fubini integration in A_i and in the third matrix gives the lower bound

$$\frac{\sqrt{(1-\vartheta)a_g b_g}}{2} \int \mathcal{S}(A_i)|\psi|^2 \, dA$$

for that summand. There are two partners j for each fixed i. Their sum and (MQ.3) prove (MQ.7), and $\vartheta = 1/2$ gives (MQ.8). Extension to (MQ.5) follows from core convergence and Fatou's lemma for the nonnegative position weight.

Here is a direct compactness proof. Fix $0 < \vartheta < 1$ and put $c_\vartheta = 2\sqrt{6(1-\vartheta)a_g b_g} > 0$. In any one retained orthonormal basis of $\mathfrak{g}^3 = \mathbb{R}^{24}$, let $Q_R = [-R,R]^{24}$. Outside this cube, $\sum_i \|A_i\| \geq (\sum_i \|A_i\|^2)^{1/2} \geq R$. Thus

$$\int_{\mathbb{R}^{24}\setminus Q_R} |\psi|^2 \leq \frac{q_{a_g,b_g}[\psi]}{c_\vartheta R}, \qquad \int |\nabla\psi|^2 \leq \frac{q_{a_g,b_g}[\psi]}{\vartheta a_g}. \tag{MQ.10}$$

Partition Q_R into finitely many cubes of side $h = 2R/N$, and let $\Pi_{R,N}\psi$ be its mean on each small cube and zero outside Q_R. This has finite-dimensional range. On any cube Q of side h,

$$\int_Q |\psi - \psi_Q|^2 \leq h^2 \int_Q |\nabla\psi|^2. \tag{MQ.11}$$

For completeness, in one dimension the variance identity gives $\int_0^h |f - f_{\mathrm{av}}|^2 = (2h)^{-1}\int_0^h\int_0^h |f(x) - f(y)|^2 \, dx\, dy$. Cauchy–Schwarz in the fundamental theorem of calculus bounds this by $h^2\int_0^h |f'|^2$. In a product cube, let P_j be averaging in the jth coordinate. The commuting orthogonal projections give the orthogonal decomposition

$$I - \prod_{j=1}^{24} P_j = \sum_{j=1}^{24} \left(\prod_{k<j} P_k \right) (I - P_j).$$

Applying the one-dimensional estimate to each summand and using that prior-coordinate averaging contracts the derivative norm proves (MQ.11), first for smooth functions and then by H^1 density.

Equations (MQ.10)–(MQ.11) give the explicit approximation bound

$$\|\psi - \Pi_{R,N}\psi\|_2^2 \le q_{a_g,b_g}[\psi]\left(\frac{h^2}{\vartheta a_g} + \frac{1}{c_\vartheta R}\right). \tag{MQ.12}$$

On a bounded form-domain set choose first R large and then N large. The error is uniformly as small as prescribed, while $\Pi_{R,N}$ maps that set to a bounded subset of a fixed finite-dimensional space. Such a set has a finite net at any positive radius, by subdividing a bounding coordinate box. Thus the original set is totally bounded in L^2, proving compactness of the form-domain inclusion without a matrix-model discreteness theorem.

The closed form defines its Friedrichs operator by the usual Riesz construction: for each $f \in L^2$ and $\mu > 0$ there is a unique u with $q_{a_g,b_g}(u,v) + \mu\langle u, v\rangle = \langle f, v\rangle$ for every form-domain v. This follows by representing the bounded functional on the Hilbert space with that form inner product; coercivity is supplied by $\mu\|v\|_2^2$. The solution is $(H_A + \mu I)^{-1}f$ and obeys $q_{a_g,b_g}[u] + \mu\|u\|_2^2 \le \|f\|_2^2/\mu$. Hence this resolvent factors through a bounded set in the form domain and its compact inclusion, so it is compact. The spectral theorem for the resulting compact self-adjoint resolvent gives finite multiplicities and eigenvalues tending to zero for the resolvent, equivalently eigenvalues of H_A tending to infinity. Since the resolvent is injective and its range is dense in an infinite-dimensional space, it has infinitely many such eigenvalues. This proves all assertions. □

The inequality is a quadratic-form inequality, not a pointwise lower bound on the quartic potential. That potential still vanishes on every commuting triple. The exact mechanism is (MQ.9): localizing the other matrices transversely to a large fixed matrix costs the nonzero oscillator ground energies in (MQ.4); their sum remains coercive even at a root wall.

Proposition 1.8 (An additional explicit lower bound on the energy origin)**.** *With precisely the energy origin in (MQ.6),*

$$H_A \ge 9(3/2)^{1/3} a_g^{2/3} b_g^{1/3}\, I. \tag{MQ.13}$$

This is a lower bound on the bottom of the operator, not a claimed lower bound of the same size on the difference between its first two energy levels.

Proof. First, for any $\alpha, \gamma > 0$,

$$-\alpha\frac{d^2}{dx^2} + \gamma|x| \ge \frac{3}{2^{8/3}}(\alpha\gamma^2)^{1/3}. \tag{MQ.14}$$

To prove it, take $f \in C_c^\infty(\mathbb{R})$ with $\|f\|_2 = 1$, and write $t = \|f'\|_2^2$ and $M = \int |x||f(x)|^2\,dx > 0$. Integration of $(|f|^2)'$ from both ends of the line gives $\|f\|_\infty^2 \le \|f\|_2\|f'\|_2 = \sqrt{t}$. Markov's inequality puts at least half of the mass inside $[-2M, 2M]$, so $1/2 \le 4M\|f\|_\infty^2 \le 4M\sqrt{t}$. Thus $tM^2 \ge 1/64$. Minimizing $\alpha/(64M^2) + \gamma M$ over $M > 0$, whose unique minimum occurs at $M^3 = \alpha/(32\gamma)$, proves (MQ.14). Homogeneity in f removes the unit-norm restriction.

Write $x_{i\nu}$ for the original metric-orthonormal real coordinates, $1 \le i \le 3$, $1 \le \nu \le 8$. Cauchy–Schwarz gives $\|A_i\| \ge 8^{-1/2}\sum_{\nu=1}^8 |x_{i\nu}|$. By (MQ.8),

$$H_A \ge \sum_{i,\nu}\left\{-\frac{a_g}{2}\partial_{x_{i\nu}}^2 + \sqrt{\frac{3a_g b_g}{2}}\,|x_{i\nu}|\right\}.$$

Apply (MQ.14) in each of these twenty-four coordinates and integrate the others. Its total constant is exactly

$$24\frac{3}{2^{8/3}}\left(\frac{a_g}{2}\,\frac{3a_g b_g}{2}\right)^{1/3} = 9(3/2)^{1/3} a_g^{2/3} b_g^{1/3}.$$

Core closure gives the operator-form bound stated. □

Exact relation to the retained homogeneous gauge density. The canonical density uses signature $(+---)$ and anti-Hermitian gauge fields, and its pure gauge part is

$$\mathcal{L}_{\text{gauge,hom}} = \frac{1}{g^2}\sum_i \|E_i\|^2 - \frac{1}{g^2}\sum_{i<j} \|[A_i, A_j]\|^2, \qquad E_i = \dot{A}_i + [A_0, A_i].$$

The full scalar potential retained there is $V = m^2 s/2 + \Gamma s^2/4 - \kappa(\det\Phi - b^*\Phi b)$, with $s = \operatorname{tr}\Phi^2 + 2b^*b$. It is not part of the pure matrix operator (MQ.6). For the pure gauge homogeneous kinetic form on a finite torus of retained volume $\mathcal{V}_3 > 0$, fixing temporal gauge at the unreduced coordinate level gives $L = \mathcal{V}_3 \sum_i \|\dot{A}_i\|^2/g^2 - \mathcal{V}_3 \sum_{i<j} \|[A_i, A_j]\|^2/g^2$. Its canonical momenta are exactly $p_i = 2\mathcal{V}_3\dot{A}_i/g^2$; the Legendre transform is

$$H_{\text{cl}} = \frac{g^2}{4\mathcal{V}_3}\sum_i \|p_i\|^2 + \frac{\mathcal{V}_3}{g^2}\sum_{i<j} \|[A_i, A_j]\|^2.$$

The specified canonical substitution $p_i = -i\hbar\nabla_{A_i}$ with $\hbar > 0$ therefore gives the exact coefficient map

$$a_g = \frac{\hbar^2 g^2}{4\mathcal{V}_3}, \qquad b_g = \frac{\mathcal{V}_3}{g^2}, \qquad a_g b_g = \frac{\hbar^2}{4}. \tag{MQ.15}$$

No coupling or volume is set to one. Under this map, (MQ.8) reads

$$H_A \geq -\frac{\hbar^2 g^2}{8\mathcal{V}_3}\Delta + \sqrt{3}\,\hbar \sum_i \|A_i\|,$$

and (MQ.13) reads $H_A \geq 9(3\hbar^4 g^2/(32\mathcal{V}_3))^{1/3} I$. These coefficient identities concern the displayed unreduced homogeneous quantum model. They impose neither the Gauss constraint nor the omitted scalar coupling, and assert no continuum or infinite-volume field-theoretic gap.

Primary literature and scope. The discreteness phenomenon for bosonic commutator matrix potentials is classical. B. Simon, *Some quantum operators with discrete spectrum but classically continuous spectrum*, Annals of Physics **146** (1983), 209–220, DOI `https://doi.org/10.1016/0003-4916(83)90057-X`, treats a compact semisimple Lie-algebra matrix operator in equation (3) and Section 7, Corollary 4. Section 2 gives the elementary zero-point-oscillator argument for the two-variable model. The present proof combines a direct slice argument with the explicit $\mathfrak{su}(3)$ root sum in the metric $-\operatorname{tr}(XY)$ and proves compactness directly by (MQ.12). It does not use the Fefferman–Phong discreteness theorem employed in that paper's Section 7, nor claim the underlying discreteness result as new.

The related finite-volume Yang–Mills analysis is M. Lüscher, *Some analytic results concerning the mass spectrum of Yang–Mills gauge theories on a torus*, Nuclear Physics B **219** (1983), 233–261, DOI `https://doi.org/10.1016/0550-3213(83)90436-4`. That field-theoretic small-volume setting is not substituted for the precisely specified matrix operator here; none of its perturbative conclusions or coefficients is assumed in the foregoing proof.

1.3 Compactness for the entire coupled Hamiltonian

Theorem 1.9 (All thirty-eight coordinates are quantum confined)**.** *Retain the configuration metric*

$$\gamma_E = \operatorname{tr}(d\Phi^2) + 2\,db^*db, \qquad E = \operatorname{Herm}_3(\mathbb{C})_0 \oplus \mathbb{C}^3,$$

and the metric $-\operatorname{tr}$ *on each of the three* $\mathfrak{su}(3)$ *factors. Let* $T_A = -\sum_i \Delta_{A_i}$ *and* $T_E = -\Delta_{\gamma_E}$*; these are nonnegative Laplacians. For every* $0 < \eta < 1$*, the full Hamiltonian* $H_{\mathrm{hom}} = H$ *obeys the quadratic-form inequality*

$$\begin{aligned} H_{\mathrm{hom}} \;\geq\; & \frac{\eta\hbar^2 g^2}{4\mathcal{V}_3} T_A + \frac{\hbar^2}{2\mathcal{V}_3} T_E + \hbar\sqrt{6(1-\eta)} \sum_{i=1}^{3} \|A_i\| \\ & + \frac{\mathcal{V}_3 \Gamma}{8} s^2 - \mathcal{V}_3 C_\kappa. \end{aligned} \tag{9}$$

Its Friedrichs realization on the entire thirty-eight-dimensional configuration space has compact resolvent. No commuting or scalar directions have been deleted in this assertion.

Proof. Apply the matrix estimate with $a_g = \hbar^2 g^2/(4\mathcal{V}_3)$ and $b_g = \mathcal{V}_3/g^2$. Their product is exactly $\hbar^2/4$. The scalar-gradient and gauge–scalar potentials

$$\frac{\mathcal{V}_3}{2} \sum_i \operatorname{tr}[A_i, \Phi]^2, \qquad \mathcal{V}_3 \sum_i \|A_i b\|^2$$

are nonnegative: $[A_i, \Phi]$ is Hermitian. Retain these terms in the definition of the operator and use their signs only in this lower bound. Lemma 1.2 now proves (9) on compactly supported smooth functions, and core closure extends it to the form domain.

Write q for the closed form and $\widetilde{q}[\psi] = q[\psi] + \mathcal{V}_3 C_\kappa \|\psi\|_2^2$. The preceding bound makes $\widetilde{q}$ nonnegative. In the original metric the squared radial coordinate is $r^2 = \sum_i \|A_i\|^2 + s$. The retained positive weight

$$W_\eta = \hbar\sqrt{6(1-\eta)} \sum_i \|A_i\| + \mathcal{V}_3 \Gamma s^2/8$$

tends to infinity as $r \to \infty$. More explicitly, outside the ball $r < R$ either $\sum_i \|A_i\|^2 \geq R^2/2$ or $s \geq R^2/2$, so

$$W_\eta \geq w_R := \min\left\{\hbar\sqrt{3(1-\eta)}R, \mathcal{V}_3 \Gamma R^4/32\right\}.$$

Both coefficients are strictly positive for the stated parameters. Thus

$$\int_{r \geq R} |\psi|^2 \leq \widetilde{q}[\psi]/w_R.$$

The derivative bound from (9) is

$$c_0 \int |\nabla\psi|^2 \leq \widetilde{q}[\psi], \qquad c_0 = \min\{\eta\hbar^2 g^2/(4\mathcal{V}_3), \hbar^2/(2\mathcal{V}_3)\} > 0,$$

where the gradient uses the product metric just stated. Metric-orthonormal coordinates are an isometric representation, not a change to any coupling, original field, or volume. On the cube $[-R, R]^{38}$, replace a function by its averages on finitely many cubes of side $h = 2R/N$ and by zero outside the large cube. The coordinate-averaging argument in the matrix proof works in thirty-eight coordinates without changing its constant: the squared local error is at most $h^2 \int |\nabla\psi|^2$. The cube's complement lies outside $r < R$, so the total error is at most

$$\widetilde{q}[\psi]\left(h^2/c_0 + 1/w_R\right).$$

On every form-norm bounded set this tends uniformly to zero by first increasing R and then increasing N. The approximations have finite-dimensional range, and their norms are bounded by

$\|\psi\|_2$. Therefore that set is totally bounded in L^2. The complete Hilbert form domain has compact inclusion in L^2. The Riesz construction with the positive form $\widetilde{q} + \|\cdot\|_2^2$ maps bounded L^2 sets to bounded form-domain sets. Its composition with this inclusion is the resolvent, proving compactness for the full operator. □

The original classical potential remains zero in its pure-gauge part for every commuting triple. Inequality (9) uses the derivative cost in directions transverse to such triples. The strictly positive coefficient in front of $\sum_i \|A_i\|$ does not assert a pointwise classical potential that was absent from the original density.

Theorem 1.10 (Strict positivity and the bottom spectral projection)**.** *For the actual polynomial Hamiltonian above and every $\tau > 0$, $e^{-\tau H}$ is positivity improving: if $0 \neq f \geq 0$, then $e^{-\tau H} f > 0$ almost everywhere. The same holds on the invariant Hilbert space. Moreover*

$$\inf \sigma(H_{\mathrm{phys}}) = \inf \sigma(H) =: E_0, \qquad \operatorname{rank} \mathbf{1}_{\{E_0\}}(H) \leq 1.$$

Every nonzero vector in that bottom eigenspace is a scalar multiple of one strictly positive normalized, $SU(3)$-invariant real function. These conclusions hold without assuming compactness.

Proof. We give the heat-kernel argument with its constants. In the above real coordinates the free kernel relative to Lebesgue measure is

$$p_\tau(q, r) = (4\pi\tau)^{-19} (\det D)^{-1/2} \exp\left(-\frac{(q-r)^{\mathsf{T}} D^{-1} (q-r)}{4\tau} \right).$$

Let B_s, $0 \leq s \leq \tau$, be a standard 38-dimensional Brownian motion and form the bridge

$$Z_s = (1 - s/\tau) q + (s/\tau) r + \sqrt{2D}\, (B_s - (s/\tau) B_\tau).$$

The Friedrichs kernel, relative to the retained measure $d\mu_M$, is

$$K_\tau(q, r) = \frac{p_\tau(q, r)}{C_M}\, \mathbb{E} \exp\left(-\int_0^\tau W(Z_s)\, ds \right). \tag{10}$$

Here τ is the semigroup parameter with inverse-energy units; Euclidean physical time is $\hbar\tau$. No value of $\hbar$ has been set to one in D.

For completeness, the formula can be justified within the present bounded-below polynomial setting. Put $W_n = \min(W + C, n)$. For bounded W_n, the Duhamel iteration of the free semigroup converges in norm, since its j-th ordered term is bounded by $(n\tau)^j / j!$. The Markov Gaussian densities of the bridge identify exactly these ordered integrals with the power-series expansion of $\mathbb{E} \exp(-\int_0^\tau W_n(Z_s)\, ds)$. Thus the kernel formula holds for the bounded potentials. The forms with potentials W_n increase to the form with potential $W + C$. Their resolvents converge strongly: for $u_n = (T + W_n + \lambda)^{-1} f$, $\lambda > 0$, the variational identities bound u_n in H^1; testing each fixed truncated form and passing weakly shows that any weak limit belongs to the limiting form domain. Compactly supported tests identify its unique limiting resolvent equation. The identities

$$\mathfrak{h}_n[u_n] + \lambda \|u_n\|^2 = \operatorname{Re}\langle f, u_n \rangle$$

and lower semicontinuity then force convergence of the L^2 norms, hence strong convergence. The spectral functional calculus gives strong convergence of their semigroups. Every bridge path is continuous on a compact interval and has bounded image. The continuous polynomial W is therefore bounded on that path, and monotone convergence of its truncated integrals and dominated

convergence of their exponentials yield (10), restoring the factor $e^{C\tau}$. This is the bounded-below Brownian-bridge Feynman–Kac formula, also stated in B. Simon, *A Feynman–Kac Formula for Unbounded Semigroups*, arXiv:math-ph/9907022, Theorem 1.1, `https://arxiv.org/abs/math-ph/9907022`.

In particular the exponential in (10) is strictly positive almost surely and bounded above by $e^{C\tau}$. Since $p_\tau(q,r) > 0$, the kernel is strictly positive for all q, r. Its Gaussian upper bound makes it a valid L^2 kernel action, and integration against any nonzero nonnegative function proves positivity improvement. The invariant subspace is reducing and the action is the same positive kernel, giving its assertion as well.

To prove equality of the two spectral infima without presupposing an eigenfunction, take any $f \in \mathscr{D}$ and define

$$F(q) = \left(\frac{1}{\mathcal{V}_\mu} \int_{SU(3)} |f(R_{g^{-1}} q)|^2 \, d\mu(g) \right)^{1/2}.$$

This is an invariant nonnegative function with $\|F\| = \|f\|$. Apply the norm inequality to the Hilbert-valued Sobolev function $q \mapsto (g \mapsto f(R_{g^{-1}} q))$. More explicitly, differentiating $(\int |f(R_{g^{-1}} q)|^2 + \varepsilon^2)^{1/2}$, using Cauchy–Schwarz, and then taking $\varepsilon \downarrow 0$, gives

$$\int (\nabla F)^* D \nabla F \, d\mu_M \le \frac{1}{\mathcal{V}_\mu} \int_{SU(3)} \int (\nabla \mathcal{U}(g) f)^* D \nabla \mathcal{U}(g) f \, d\mu_M \, d\mu(g).$$

One first argues locally for smooth functions and then uses form approximation and lower semicontinuity. Invariance of W gives equality of the potential integrals. Thus $F \in \mathscr{D}$ and $\mathfrak{h}[F] \le \mathfrak{h}[f]$. The variational characterization of the spectral infimum proves $\inf \sigma(H_{\text{phys}}) \le \inf \sigma(H)$; the reverse inequality follows from restriction to a subspace.

Finally suppose the bottom spectral projection is nonzero, and take a real nonzero bottom eigenfunction u; real and imaginary parts are eigenfunctions because H is real. The weak-gradient inequality $|\nabla |u|| \le |\nabla u|$ gives $\mathfrak{h}[|u|] \le E_0 \|u\|^2$. The lower spectral bound forces equality, and the spectral theorem then gives $|u| \in \ker(H - E_0)$. Positivity improvement makes $|u| > 0$ almost everywhere. A sign-changing real bottom eigenfunction is impossible: its positive and negative parts are both sent to strictly positive functions, so

$$|e^{-\tau H} u| < e^{-\tau H} |u|$$

almost everywhere, contradicting the eigenfunction identities. Two independent real bottom eigenfunctions would yield, by orthogonalization against the positive one, such a sign-changing eigenfunction. Thus the complex eigenspace has dimension one. Every $\mathcal{U}(g)|u|$ is another positive normalized bottom eigenfunction, so its one-dimensional proportionality constant is positive and of modulus one, hence one. This proves gauge invariance and all assertions of the theorem. □

Corollary 1.11 (Unique physical ground state and a finite-model gap)**.** *The actual homogeneous quantum Hamiltonian has one normalized positive ground state, unique up to phase and invariant under spatially constant $SU(3)$. The physical eigenvalues, counted with multiplicity, satisfy*

$$E_0 = E_0^{\text{phys}} < E_1^{\text{phys}} \le E_2^{\text{phys}} \le \cdots, \qquad E_n^{\text{phys}} \longrightarrow +\infty.$$

In particular $E_1^{\text{phys}} - E_0 > 0$, without a numerical value being claimed.

Proof. The positive compact resolvent $(H + C + 1)^{-1}$ proved above has a nonzero top eigenvalue, whose inverse supplies an eigenfunction at the bottom of H. The strict-positivity theorem makes

it unique and invariant. The physical space is infinite dimensional, and its compact resolvent has finite-multiplicity eigenvalues accumulating only at zero. Applying the inverse affine spectral relation proves the displayed sequence and its strict first inequality. □

This is a gap of the specified finite-dimensional quantum mechanics. No numerical value follows from positivity, and classical slow solutions do not compute its level spacings. The proof retains fixed positive $\hbar, g^2, \mathcal{V}_3, \Gamma$; it asserts no uniform gap in limits of these parameters or in a continuum limit.

Proposition 1.12 (Exact classical embedding and its quotient limitations)**.** *Let ι send each homogeneous configuration and velocity to the corresponding spatially constant fields on the trivial $SU(3)$ bundle over $\mathbb{T}^3$. The full action pulls back to* (1)*. Its homogeneous Euler–Lagrange and Gauss solutions are exact full classical solutions. On phase space use constant momentum densities $\Pi_i/\mathcal{V}_3, \Pi_\Phi/\mathcal{V}_3, \pi_b/\mathcal{V}_3$; then the full canonical one-form pulls back to ϑ exactly. The map is equivariant for spatially constant $SU(3)$. The induced map from its constant-group orbit space to the full torus gauge orbit space is not injective.*

Proof. Spatial derivatives of the retained coefficients vanish. Substitution in the density gives (1), and the full local Euler–Lagrange residuals are then constant in space. Vanishing of their homogeneous variations for every time-dependent matrix variation therefore makes those residuals zero pointwise, including the A_0 residual. They consequently vanish against arbitrary, not necessarily homogeneous, spacetime test variations. This proves the exact classical implication without a symmetry reduction heuristic. Integrating constant momentum densities against constant configuration variations multiplies them by $\mathcal{V}_3$, proving the asserted one-form pullback and thus its symplectic pullback. Constant-group equivariance is direct.

For the limitation, let L_1 be the first torus period and set

$$g(x_1) = \operatorname{diag}(e^{2\pi i x_1/L_1}, e^{-2\pi i x_1/L_1}, 1).$$

With $D = \partial + A$, gauge transformation is

$$A_i' = gA_ig^{-1} - (\partial_i g)g^{-1}, \quad \Phi' = g\Phi g^*, \quad b' = gb.$$

The homogeneous state $A_i = 0, \Phi = b = 0$ is thereby equivalent in the full torus theory to the homogeneous state

$$A_1' = -\frac{2\pi i}{L_1}\operatorname{diag}(1,-1,0), \qquad A_2' = A_3' = 0, \qquad \Phi' = b' = 0.$$

No constant conjugation sends the zero A_1 to this nonzero matrix, proving noninjectivity of the orbit-space map. On the other hand the same spatial gauge function sends the constant $\Phi = E_{12} + E_{21}$ to

$$e^{4\pi i x_1/L_1}E_{12} + e^{-4\pi i x_1/L_1}E_{21},$$

which is not homogeneous. Thus spatial gauge maps do not act on the whole homogeneous ansatz. More generally, a gauge map preserving every homogeneous triplet $b = e_j$, $j = 1, 2, 3$, must have all three columns constant, and hence must itself be spatially constant. This argument applies in particular to any genuinely large transformation not homotopic to a constant. The explicit diagonal example above is not being mislabeled as topologically large in $SU(3)$; its periodicity alone already proves the required quotient obstruction. □

Why the classical embedding is not a quantum-sector theorem. There is a direct interaction obstruction already in a finite Fourier cutoff. Choose the trace-free Hermitian $S = \mathrm{diag}(1, -1, 0)/\sqrt{2}$, so $\mathrm{tr}\, S^2 = 1$, $\det S = 0$, and put

$$\Phi(x) = (q + y\cos(2\pi x_1/L_1))S, \qquad A_i = b = 0.$$

The quartic part of the spatially integrated potential is exactly

$$\frac{\Gamma\mathcal{V}_3}{4}\left(q^4 + 3q^2y^2 + \frac{3}{8}y^4\right).$$

Indeed the averages of the cosine, its square, its cube and its fourth power are $0, 1/2, 0, 3/8$. The mass and gradient terms separate in q, y, the cubic determinant vanishes, but the mixed term $3\Gamma\mathcal{V}_3 q^2y^2/4$ remains. The kinetic density is $\mathcal{V}_3\dot{q}^2/2 + \mathcal{V}_3\dot{y}^2/4$; thus this example also retains the unequal Fourier-mode masses.

A putative product sector consisting of all q-wavefunctions times one fixed nonzero normalizable y-wavefunction Ω is not invariant under this interacting cutoff Hamiltonian. Here invariance means a reducing quantum subspace, equivalently that its orthogonal projection commutes with the unitary evolution. This includes the necessary domain issue: if that projection commuted with the Hamiltonian, applying it to $f(q)v(y) \in C_c^\infty$, with $\langle\Omega, v\rangle \neq 0$, would put $f(q)\Omega(y)$ in the operator domain for every $f \in C_c^\infty(\mathbb{R})$. Its distributional Hamiltonian is the separated q-term plus $f(q)H_y\Omega + cq^2f(q)y^2\Omega$, $c = 3\Gamma\mathcal{V}_3/4$. Integrating against two compact q-test functions with distinct averaged q^2 values isolates $H_y\Omega$ and $y^2\Omega$ as L^2 functions. Thus the following projections are legitimate even if no regularity of Ω was assumed initially. Projecting its action orthogonally to Ω, for arbitrary compact smooth q-wavefunctions, would require

$$(H_y\Omega)_\perp + \frac{3\Gamma\mathcal{V}_3}{4}q^2(y^2\Omega)_\perp = 0 \quad \text{for almost every } q.$$

Its two coefficients must vanish, implying $y^2\Omega = \lambda_\Omega\Omega$. Multiplication by y^2 has no nonzero $L^2(\mathbb{R})$ eigenfunction: its level sets have Lebesgue measure zero. Thus no fixed discarded-mode vacuum yields that product invariant sector. The distribution $\delta(y)$, which would impose a literally vanishing discarded coordinate, is not an L^2 state either. This does not rule out every possible correspondence with a full quantum theory; it disproves the specific automatic-subsector inference and states exactly which finite model has been quantized. No continuum quantum Hilbert space, full local-gauge quotient, renormalized Hamiltonian, or continuum mass-gap value is constructed by the finite-dimensional argument.

Gauss-law reference. D. Tong, *Lectures on Gauge Theory*, Section 2.2.1, equations (2.27)–(2.31), `https://www.damtp.cam.ac.uk/user/tong/gaugetheory.html`, retains the Gauss equation after temporal gauge and imposes it on physical states. His Hermitian-field normalization is not substituted for the retained anti-Hermitian normalization here; the exact factors and the sign of (3) were derived above from the given density. For this compact spatial torus, spatially constant $SU(3)$ is the finite gauge subgroup being imposed, rather than an asymptotic charge group at spatial infinity.

2 An exact moving-ground-state comparison for the retained Fourier pair

The result below concerns precisely the two-coordinate variational truncation

$$\Phi(x) = (q + y\cos(kx_1))S, \qquad S = \mathrm{diag}(1, -1, 0)/\sqrt{2}, \qquad k = 2\pi/L_1, \qquad A_i = b = 0.$$

It does not assume that this two-dimensional plane is an invariant subspace of the full classical field equations or a reducing subspace of an independently constructed full quantum field theory. Throughout, retain

$$\hbar > 0, \quad \mathcal{V}_3 > 0, \quad \Gamma > 0, \quad L_1 > 0, \qquad m^2 \geq 0,$$
$$\omega_* := \sqrt{m^2 + k^2} > 0, \qquad \omega(q) := \sqrt{\omega_*^2 + 3\Gamma q^2}.$$

All bounds will refer to these original parameters and coordinates. The quartic discarded-mode interaction is never removed from the operator.

2.1 Exact coefficients, measures, and self-adjoint operators

Since $\operatorname{tr} S^2 = 1$ and $\det S = 0$, the pullback of the cubic determinant is zero. The averages of $\cos(kx_1)$ and its first four powers are $0, 1/2, 0, 3/8$ for powers one through four. The exact integrated Lagrangian is therefore

$$L_2 = \frac{\mathcal{V}_3}{2}\dot{q}^2 + \frac{\mathcal{V}_3}{4}\dot{y}^2 - V_q(q) - \frac{\mathcal{V}_3\omega(q)^2}{4}y^2 - cy^4, \tag{11}$$
$$V_q(q) := \mathcal{V}_3\left(\frac{m^2q^2}{2} + \frac{\Gamma q^4}{4}\right), \qquad c := \frac{3\Gamma\mathcal{V}_3}{32} > 0.$$

In particular the gradient term is $\mathcal{V}_3k^2y^2/4$ and the mixed quartic term is $3\Gamma\mathcal{V}_3q^2y^2/4$. The momenta are $p_q = \mathcal{V}_3\dot{q}$ and $p_y = \mathcal{V}_3\dot{y}/2$. Thus the mass metric and its measures are

$$M_2 = \operatorname{diag}(\mathcal{V}_3, \mathcal{V}_3/2), \quad d\mu_q = \sqrt{\mathcal{V}_3}\,dq, \quad d\mu_y = \sqrt{\mathcal{V}_3/2}\,dy, \quad d\mu_2 = \frac{\mathcal{V}_3}{\sqrt{2}}\,dq\,dy = d\mu_q d\mu_y.$$

We retain these constants. Put

$$\mathscr{K} = L^2(\mathbb{R}, d\mu_y), \qquad \mathscr{L} = L^2(\mathbb{R}, d\mu_q), \qquad \mathscr{H}_2 = L^2(\mathbb{R}, d\mu_q; \mathscr{K}).$$
$$a := \frac{\hbar^2}{2\mathcal{V}_3}, \quad a_y := \frac{\hbar^2}{\mathcal{V}_3}, \quad b_q := \frac{\mathcal{V}_3\omega(q)^2}{4}.$$

The Hamiltonian is exactly

$$H_2 = -a\partial_q^2 + V_q(q) + h(q), \qquad h(q) = -a_y\partial_y^2 + b_qy^2 + cy^4. \tag{12}$$

The symbols a, a_y, b_q, c denote these retained coefficients, not a rescaling of either variable.

Every displayed nonnegative Schrödinger expression is realized by its Friedrichs form. For example

$$\mathcal{D}_y = H^1(\mathbb{R}, d\mu_y) \cap \{u : y^2u \in \mathscr{K}\}, \qquad h_q[u] = a_y\|u'\|^2 + b_q\|yu\|^2 + c\|y^2u\|^2.$$

Its form domain is independent of q, since $b_q, c > 0$. The full form domain consists of $\Psi \in H^1(\mathbb{R}^2, d\mu_2)$ for which the original nonnegative multiplication potential in (12) has finite integral against $|\Psi|^2$. Completeness follows by convergence in H^1 and after multiplication by the square root of that potential; an almost-everywhere subsequence identifies the limits. Smooth cutoffs and then compactly supported mollification prove that C_c^∞ is a form core. The operator domain is the form-domain vectors whose displayed differential expression belongs to L^2 distributionally, as follows by testing against the core and using the Riesz representation of the shifted form. Since $b_qy^2 + cy^4 \to \infty$ in y, the fibre form-domain inclusion is compact: the potential controls the L^2 tail and the derivative

controls finite-cell averaging errors on bounded intervals. The same argument proves compactness for H_2, using $V_q(q) + cy^4 \to \infty$ as $q^2 + y^2 \to \infty$.

We shall also use the following elementary operator-core fact. For a smooth nonnegative real potential on $\mathbb{R}^n$ and a constant positive kinetic matrix, C_c^∞ is an operator core for the Friedrichs operator. Indeed an L^2 distributional solution of $(H^* + 1)u = 0$ is locally H^2 by local elliptic regularity. Testing against $\chi_R^2 u$, with a smooth compact cutoff whose gradient is $O(R^{-1})$, gives

$$\int (|\nabla(\chi_R u)|_D^2 + (V+1)|\chi_R u|^2) = \int |\nabla \chi_R|_D^2 |u|^2 = O(R^{-2})\|u\|^2.$$

Letting $R \to \infty$ proves $u = 0$. Thus the range of $H + 1$ on the minimal domain is dense; its lower bound makes the range of its closure closed. That closure is surjective at -1 and symmetric, hence self-adjoint. It is therefore the Friedrichs realization and the stated domain is an operator core. This argument applies to the fibres, the full operator, and the effective operator below.

2.2 The actual normalized positive fibre ground state

Lemma 2.1 (Fibre energies and a proved quantitative separation)**.** *For every q, $h(q)$ has a simple lowest eigenvalue $e(q)$ and a unique real strictly positive eigenfunction Ω_q with $\|\Omega_q\|_{\mathscr{K}} = 1$. Write $e_1(q)$ for its second eigenvalue, counting multiplicity. Then*

$$\frac{\hbar\omega(q)}{2} \le e(q) \le \frac{\hbar\omega(q)}{2} + \frac{9\Gamma\hbar^2}{32\mathcal{V}_3\omega(q)^2}, \qquad e_1(q) \ge \frac{3\hbar\omega(q)}{2}. \tag{13}$$

In the explicit parameter regime

$$\frac{9\Gamma\hbar}{32\mathcal{V}_3\omega_*^3} \le \frac{1}{2}, \quad \textit{equivalently} \quad \mathcal{V}_3 \ge \frac{9\Gamma\hbar}{16\omega_*^3}, \tag{14}$$

these imply, simultaneously for every q,

$$e(q) \le \hbar\omega(q), \qquad e_1(q) - e(q) \ge \lambda(q) := \frac{\hbar\omega(q)}{2} > 0. \tag{15}$$

Proof. Compactness gives existence of a bottom eigenfunction. The scalar kinetic form does not increase on taking its modulus, so a nonnegative real eigenfunction can be selected. Alternatively, the continuous nonnegative potential has the Brownian-bridge kernel

$$\frac{\exp[-(y-z)^2/(4a_y\tau)]}{\sqrt{\mathcal{V}_3/2}\sqrt{4\pi a_y\tau}}\mathbb{E}\exp\left[-\int_0^\tau (b_q Z_s^2 + cZ_s^4)\,ds\right] > 0 \qquad (\tau > 0).$$

Here Z is the bridge from y to z with diffusion covariance $2a_y$. Each path is continuous with bounded image, so its exponential is strictly positive and bounded by one. The formula follows first for bounded truncations by the convergent Duhamel series and then by increasing-form resolvent convergence and dominated convergence of the bridge exponential. Consequently the semigroup is positivity improving. The modulus of any bottom eigenfunction is again a bottom eigenfunction; if a real bottom eigenfunction changed sign, positivity improvement on both its positive and negative parts would imply $|e^{-\tau h(q)}u| < e^{-\tau h(q)}|u|$, contradicting its eigenvalue equations. Orthogonalization against a positive bottom eigenfunction therefore rules out a second independent real one. Reality gives complex dimension one. The eigenfunction is smooth by the one-dimensional eigenvalue equation. A zero of a nonnegative smooth solution would have derivative zero, and uniqueness

for that second-order ODE would force the solution to vanish. Thus the selected eigenfunction is strictly positive everywhere.

Put $\beta(q) = \mathcal{V}_3\omega(q)/(2\hbar)$ and use the exactly normalized harmonic comparison vector

$$\chi_q(y) = \big(\mathcal{V}_3/2\big)^{-1/4}\big(\beta(q)/\pi\big)^{1/4}\exp[-\beta(q)y^2/2], \qquad \|\chi_q\|_{\mathscr{K}} = 1.$$

The harmonic part $h_{\mathrm{osc}}(q) = -a_y\partial_y^2 + b_q y^2$ factorizes as

$$h_{\mathrm{osc}}(q) = a_y(-\partial_y + \beta y)(\partial_y + \beta y) + \hbar\omega/2.$$

Its ground state is χ_q and its first excitation has energy $3\hbar\omega/2$. To check the latter without presuming a missing spectral level, the harmonic operator has compact resolvent; its lowering operator $\partial_y + \beta y$ maps any eigenfunction above the ground to an L^2 distributional eigenfunction with energy lowered by $\hbar\omega$. Its norm is nonzero by the factorization. No energy strictly between $\hbar\omega/2$ and $3\hbar\omega/2$ is possible. The function $y\chi_q$ attains the latter energy. The operator-core fact above identifies the distributional eigenfunctions with the self-adjoint operator domain.

Since $cy^4 \geq 0$, the variational principle and its two-dimensional min–max version give the lower bounds for e and e_1. Gaussian integration gives, in the retained measure,

$$\langle\chi_q, y^2\chi_q\rangle = \frac{\hbar}{\mathcal{V}_3\omega}, \qquad \langle\chi_q, y^4\chi_q\rangle = \frac{3\hbar^2}{\mathcal{V}_3^2\omega^2}.$$

Its Rayleigh quotient is the upper bound in (13). This is only a trial-vector bound: $h(q)\chi_q = (\hbar\omega/2 + cy^4)\chi_q$ is not proportional to χ_q because $c > 0$. In particular the true state Ω_q is not replaced by this Gaussian. Finally $\omega(q) \geq \omega_*$ and (14) make the upper correction at most $\hbar\omega(q)/2$, proving (15). □

Lemma 2.2 (An exact smooth eigenline, with its phase fixed)**.** *The functions $q \mapsto e(q)$ and $q \mapsto \Omega_q \in \mathscr{K}$ are locally real analytic, the latter also in the common local operator graph norm. Thus they define a global normalized positive ground-state line over the entire original q-axis. In this phase,*

$$\langle\Omega_q, \partial_q\Omega_q\rangle = 0, \qquad \mathcal{A}(q) := i\langle\Omega_q, \partial_q\Omega_q\rangle = 0.$$

Put $P_q = |\Omega_q\rangle\langle\Omega_q|$, $Q_q = I - P_q$, and

$$R_q = (h(q) - e(q))^{-1}Q_q, \qquad d := \frac{3\Gamma\mathcal{V}_3}{2}.$$

The inverse is on the orthogonal complement of the actual ground state. The exact derivative identities are

$$\begin{aligned} e'(q) &= dq\,\langle\Omega_q, y^2\Omega_q\rangle, \\ \Omega_q' &= -dq\,R_qQ_qy^2\Omega_q, \\ Q_q\Omega_q'' &= -R_qQ_q[dy^2\Omega_q + 2(dqy^2 - e'(q))\Omega_q'], \\ \langle\Omega_q, \Omega_q''\rangle &= -\|\Omega_q'\|^2. \end{aligned} \tag{16}$$

These identities hold without imposing (14)*.*

Proof. For u in the operator domain of a fixed fibre $h(q_0)$,

$$c\|y^2u\|^2 \leq \langle u, h(q_0)u\rangle \leq \|u\|\,\|h(q_0)u\|.$$

The arithmetic–geometric mean inequality implies that multiplication by y^2 is $h(q_0)$-bounded with arbitrarily small relative bound. Since $h(q) - h(q_0) = (b_q - b_{q_0})y^2$, nearby real fibres have the same operator domain. One can see the corresponding resolvent analyticity directly: at a fixed resolvent point z,

$$(h(q) - z)^{-1} = (h(q_0) - z)^{-1}\big[I + (b_q - b_{q_0})y^2(h(q_0) - z)^{-1}\big]^{-1},$$

where the last inverse is a locally convergent Neumann series. For real q, the closed symmetric perturbation is self-adjoint: the same formula gives surjectivity at two conjugate nonreal points chosen far enough from the real axis. It agrees with the Friedrichs realization on the compact smooth operator core. The displayed formula is analytic also as a map into the original operator domain equipped with its graph norm.

A small contour around the isolated simple eigenvalue $e(q_0)$ defines the spectral projection by the resolvent integral. It is locally analytic in that graph norm, remains of rank one because close projections have isomorphic ranges, and gives the locally analytic eigenvalue. For a fully fixed positive reference vector use $\chi_*(y) = \chi_0(y)$, the preceding Gaussian at ω_*. Then the actual state is exactly

$$\Omega_q = \frac{P_q\chi_*}{\|P_q\chi_*\|_{\mathscr{K}}}. \tag{17}$$

The denominator is nonzero because the ground state and χ_* are both strictly positive. This formula selects the same real positive vector on overlaps of the local contour constructions, so the local analytic sections give a global section. It can equivalently be obtained by the convergent normalized heat iteration $e^{-\tau h(q)}\chi_*/\|e^{-\tau h(q)}\chi_*\|$ as $\tau \to \infty$, by the discrete spectral expansion and the nonzero ground-state overlap.

The real normalization identity gives $\langle \Omega_q, \Omega_q' \rangle = 0$. Differentiating the eigenvalue equation in its common graph domain gives $(h - e)\Omega' = -(dqy^2 - e')\Omega$. Taking its ground-state component proves the first identity, and taking its complementary component proves the second. A second differentiation gives $(h - e)\Omega'' = -(dy^2 - e'')\Omega - 2(dqy^2 - e')\Omega'$. Its complementary part gives the asserted reduced-resolvent formula; differentiating the normalization twice gives its ground-state component. All products in these identities are in $\mathscr{K}$, since y^2 is bounded in the local graph norm. □

2.3 Explicit global derivative bounds for the true quartic state

Lemma 2.3 (Moment and reduced-resolvent estimates)**.** *Assume the proved parameter regime* (14)*. Then, globally in q,*

$$\|\Omega_q'\| \le M_1 := \frac{\sqrt{15\Gamma}}{\omega_*}, \tag{18}$$

$$\|Q_q\Omega_q''\| \le M_2 := \frac{18\sqrt{5}\,\Gamma}{\omega_*^2} + \frac{192}{5^{3/4}}\sqrt{\frac{\Gamma\mathcal{V}_3}{\hbar\omega_*}}. \tag{19}$$

Proof. In this proof all fibre inner products use the retained measure $d\mu_y$. Let $\mu_j = \langle \Omega_q, y^j\Omega_q \rangle$ for even j. The energy identity gives $b_q\mu_2 \le e(q)$. To get the necessary sharper fourth moment, take the bounded Lipschitz multiplier $F_R(y) = \max(-R, \min(y, R))$. Testing the weak eigenvalue equation with $F_R^2\Omega_q$ gives the exact identity

$$a_y\|(F_R\Omega_q)'\|^2 + \int (b_q y^2 + cy^4)F_R^2|\Omega_q|^2\,d\mu_y = e(q)\|F_R\Omega_q\|^2 + a_y\int |F_R'|^2|\Omega_q|^2\,d\mu_y.$$

The test lies in the form domain because F_R and its derivative are bounded. Since $|F_R| \le |y|$, $|F_R'| \le 1$, dropping nonnegative terms and letting $R \to \infty$ by Fatou yields

$$b_q \mu_4 \le e(q)\mu_2 + a_y, \qquad \mu_4 \le \frac{e(q)^2}{b_q^2} + \frac{a_y}{b_q}.$$

By (15), this gives the explicit estimates

$$\mu_2 \le \frac{4\hbar}{\mathcal{V}_3 \omega(q)}, \qquad \|y^2 \Omega_q\| = \sqrt{\mu_4} \le \frac{2\sqrt{5}\,\hbar}{\mathcal{V}_3 \omega(q)}. \tag{20}$$

No Gaussian moment has been assigned to the quartic ground state.

From the proved gap, $\|R_q\| \le 1/\lambda(q)$. If $v \in \mathscr{K}$ and $w = R_q Q_q v$, the exact equation $(h-e)w = Q_q v$ and $w \perp \Omega_q$ give

$$c\|y^2 w\|^2 \le \langle w, h(q) w\rangle = e(q)\|w\|^2 + \mathrm{Re}\langle w, v\rangle \le \frac{e(q)+\lambda(q)}{\lambda(q)^2}\|v\|^2.$$

Thus (16) implies the pointwise estimates

$$\begin{aligned} \|\Omega_q'\| &\le L(q) := \frac{6\sqrt{5}\,\Gamma|q|}{\omega(q)^2}, \\ \|y^2 \Omega_q'\| &\le L(q)\sqrt{\frac{e(q)+\lambda(q)}{c}} \le 4L(q)\sqrt{\frac{\hbar\omega(q)}{\Gamma \mathcal{V}_3}}, \\ |e'(q)| &\le \frac{6\Gamma\hbar|q|}{\omega(q)}. \end{aligned} \tag{21}$$

Substitution in the exact second-derivative identity gives

$$\|Q_q \Omega_q''\| \le \frac{6\sqrt{5}\,\Gamma}{\omega(q)^2} + 144\sqrt{5}\,\Gamma^{3/2}\sqrt{\frac{\mathcal{V}_3}{\hbar}}\frac{q^2}{\omega(q)^{5/2}} + 144\sqrt{5}\,\Gamma^2 \frac{q^2}{\omega(q)^4}. \tag{22}$$

For completeness the exact scalar maxima used to make these bounds global are

$$\begin{aligned} \sup_q \frac{|q|}{\omega_*^2 + 3\Gamma q^2} &= \frac{1}{2\sqrt{3\Gamma}\,\omega_*}, \\ \sup_q \frac{q^2}{(\omega_*^2 + 3\Gamma q^2)^2} &= \frac{1}{12\Gamma\omega_*^2}, \\ \sup_q \frac{q^2}{(\omega_*^2 + 3\Gamma q^2)^{5/4}} &= \frac{4}{3\Gamma\, 5^{5/4}\sqrt{\omega_*}}. \end{aligned}$$

Differentiation in $|q|$ or q^2 shows that the maxima occur, respectively, at $q^2 = \omega_*^2/(3\Gamma)$, $q^2 = \omega_*^2/(3\Gamma)$, and $q^2 = 4\omega_*^2/(3\Gamma)$; each function decreases to zero at infinity. Applying these maxima to (21) and (22) gives (18) and (19), including the constants $18\sqrt{5}$ and $192/5^{3/4}$. □

2.4 The exact projected Hamiltonian and its off-diagonal part

Theorem 2.4 (Exact isometric correspondence, not an invariant product)**.** *For all of the retained positive parameters, without the extra restriction* (14)*, define the isometry*

$$J : \mathscr{L} \longrightarrow \mathscr{H}_2, \qquad (Jf)(q,y) = f(q)\Omega_q(y), \qquad (J^*\Psi)(q) = \langle \Omega_q, \Psi(q,\cdot)\rangle_{\mathscr{K}}.$$

Its range projection is the fibrewise projection $(\mathsf{P}\Psi)(q) = P_q\Psi(q)$*, and* $\mathsf{Q} = I - \mathsf{P}$*. The exact pullback of the full quadratic form is the Friedrichs form of*

$$H_{\text{eff}} = -\frac{\hbar^2}{2\mathcal{V}_3}\partial_q^2 + V_q(q) + e(q) + B(q), \qquad B(q) = \frac{\hbar^2}{2\mathcal{V}_3}\|\Omega_q'\|_{\mathscr{K}}^2. \tag{23}$$

Its domain is precisely

$$\mathcal{D}_{\text{eff}} = \left\{ f \in H^1(\mathbb{R}, d\mu_q) : \int (V_q + e + B)|f|^2 \, d\mu_q < \infty \right\},$$

and $J^{-1}(\mathcal{D}(H_2^{1/2}) \cap \operatorname{Ran} J) = \mathcal{D}_{\text{eff}}$*. The Berry connection is zero in the fixed positive real phase, but* $B(0) = 0$ *and* $B(q) > 0$ *for every* $q \neq 0$*.*

For every $f \in C_c^\infty(\mathbb{R})$*,* $Jf \in D(H_2)$ *and the exact operator identity is*

$$\begin{aligned} H_2 Jf &= J H_{\text{eff}} f + \mathcal{R} f, \\ (\mathcal{R}f)(q) &= -a\big(2f'(q)\Omega_q' + f(q) Q_q \Omega_q''\big), \qquad \mathsf{P}\mathcal{R} f = 0. \end{aligned} \tag{24}$$

Thus $J^* H_2 J = H_{\text{eff}}$ *on this core, while* $\mathsf{Q} H_2 J = \mathcal{R}$ *is the retained off-diagonal residual. The range of* J *is not a reducing subspace.*

Under (14)*, these identities extend to all* $f \in D(H_{\text{eff}})$*, and*

$$\|\mathcal{R} f\| \le 2\sqrt{a}\, M_1 \|H_{\text{eff}}^{1/2} f\| + a M_2 \|f\|, \qquad 0 \le B(q) \le \frac{15\Gamma\hbar^2}{2\mathcal{V}_3 \omega_*^2}. \tag{25}$$

Proof. Normalization of the actual fibre state and Fubini give $\|Jf\| = \|f\|$ and the displayed adjoint and projection. For compact smooth f,

$$\partial_q(Jf) = f'\Omega_q + f\Omega_q', \qquad \|\partial_q(Jf)(q)\|^2 = |f'|^2 + |f|^2\|\Omega_q'\|^2,$$

because $\langle \Omega_q, \Omega_q' \rangle = 0$. The fibre energy is exactly $e(q)|f(q)|^2$, not a harmonic replacement. These identities prove equality of the pulled-back forms on the core. Core approximation gives the forward domain inclusion. Conversely, if Jf lies in the full form domain, local Hilbert-valued Sobolev differentiation gives $f' = \langle \Omega_q, \partial_q(Jf) \rangle$ distributionally; the term with Ω_q' vanishes by normalization. Hence $f \in H^1$, and the displayed nonnegative derivative and fibre-energy identities give every required integral in $\mathcal{D}_{\text{eff}}$. This proves equality of the domains and closed forms. The positive smooth effective potential defines its self-adjoint Friedrichs operator with the operator core proved above. It has compact resolvent because it is bounded below by $V_q(q)$, which tends to infinity, and its kinetic coefficient a is strictly positive.

By local graph regularity of Ω_q, Jf is in the full form domain and its distributional Hamiltonian is in $\mathscr{H}_2$. It therefore lies in $D(H_2)$. Expanding $\partial_q^2(f\Omega_q)$ and using $\langle \Omega_q, \Omega_q'' \rangle = -\|\Omega_q'\|^2$ proves (24) and the orthogonality of its residual.

At $q = 0$, (16) gives $\Omega_0' = 0$. If $q \neq 0$ and $\Omega_q' = 0$, that identity and injectivity of the reduced inverse on $Q_q\mathscr{K}$ would give $Q_q y^2 \Omega_q = 0$. It would follow that $y^2\Omega_q$ is a scalar multiple of Ω_q. Multiplication by y^2 has no nonzero L^2 eigenfunction, because every level set has measure zero. Thus $\Omega_q' \neq 0$ for $q \neq 0$ and $B(q) > 0$ there. If the range of J were reducing, its residual would vanish for every compact smooth f. Since its coefficients are continuous in q, choose at any $q_0 \neq 0$ a test function with $f(q_0) = 0$, $f'(q_0) = 1$. The residual at q_0 is then $-2a\Omega_{q_0}' \neq 0$, a contradiction.

Under (14), the global derivative estimates and the triangle inequality give

$$\|\mathcal{R} f\| \le a(2M_1\|f'\| + M_2\|f\|).$$

All terms in the effective potential are nonnegative, so $\sqrt{a}\|f'\| \le \|H_{\rm eff}^{1/2} f\|$. This proves (25) and defines $\mathcal{R}$ as a bounded map from the effective form domain with its form norm to $\mathscr{H}_2$. For $f \in D(H_{\rm eff})$, choose an operator-core sequence. It converges also in form norm by the spectral inequality $\|H_{\rm eff}^{1/2} u\|^2 \le \|u\| \|H_{\rm eff} u\|$. The right side of (24) then converges in $\mathscr{H}_2$. Closedness of H_2 shows that $Jf \in D(H_2)$ and proves the identity on the full effective operator domain. The bound on B follows directly from aM_1^2. □

2.5 Controlled physical-time and spectral comparisons

Theorem 2.5 (Explicit finite-energy error in a proved parameter regime)**.** *Assume* (14)*. Let* $f \in \mathcal{D}_{\rm eff}$ *have* $\|f\| = 1$ *and effective energy expectation* $\|H_{\rm eff}^{1/2} f\|^2 \le E$*. For every physical time* $t \in \mathbb{R}$*,*

$$\left\| e^{-itH_2/\hbar} Jf - J e^{-itH_{\rm eff}/\hbar} f \right\| \le |t|\, \mathcal{E}(E), \tag{26}$$

$$\mathcal{E}(E) := \frac{\sqrt{30\Gamma E}}{\omega_* \sqrt{\mathcal{V}_3}} + \frac{9\sqrt{5}\,\Gamma\hbar}{\mathcal{V}_3 \omega_*^2} + \frac{96}{5^{3/4}} \sqrt{\frac{\Gamma\hbar}{\mathcal{V}_3 \omega_*}}.$$

The same bound holds for the leakage norm $\|\mathsf{Q} e^{-itH_2/\hbar} Jf\|$*. In particular, for any prescribed* $T > 0$*,* $0 < \rho < 1$*, the entirely explicit sufficient volume regime*

$$\mathcal{V}_3 \ge \max\left\{ \frac{9\Gamma\hbar}{16\omega_*^3},\ \frac{270\Gamma E T^2}{\rho^2 \omega_*^2},\ \frac{27\sqrt{5}\,\Gamma\hbar T}{\rho\omega_*^2},\ \frac{288^2 \Gamma\hbar T^2}{5^{3/2}\rho^2\omega_*} \right\} \tag{27}$$

makes both norms at most ρ *simultaneously for* $|t| \le T$*. All compared Hamiltonians still contain the original quartic interaction, and no coupling has been set to one.*

Proof. First take $f \in D(H_{\rm eff})$. The established operator-domain map permits differentiation of $e^{-i(t-s)H_2/\hbar} J e^{-isH_{\rm eff}/\hbar} f$. Integrating its derivative gives the exact Duhamel identity

$$e^{-itH_2/\hbar} Jf - J e^{-itH_{\rm eff}/\hbar} f = -\frac{i}{\hbar} \int_0^t e^{-i(t-s)H_2/\hbar} \mathcal{R} e^{-isH_{\rm eff}/\hbar} f\, ds.$$

Unitarity and preservation of the effective form norm give the bound $|t|(2\sqrt{a}M_1\sqrt{E} + aM_2)/\hbar$. Substitution of the exact a, M_1, M_2 yields (26). For a general form-domain vector, effective spectral cutoffs converge both in norm and form norm and give the same estimate by continuity. The comparison vector on the right lies in $\operatorname{Ran} J$, so applying the contraction Q proves the leakage bound. The last three lower bounds on volume in (27) make respectively the three terms of $T\mathcal{E}(E)$ at most $\rho/3$; the first is the separately proved fibre-gap regime. Squaring the first and third nonnegative inequalities and directly solving the second gives exactly their displayed constants. □

Corollary 2.6 (An explicit approximate-eigenstate relation)**.** *In* (14)*, every normalized effective eigenvector* $H_{\rm eff} f = \Lambda f$ *satisfies*

$$\|(H_2 - \Lambda) Jf\| \le 2\sqrt{a} M_1 \sqrt{\Lambda} + aM_2 = \hbar\mathcal{E}(\Lambda), \qquad \operatorname{dist}(\Lambda, \sigma(H_2)) \le \hbar\mathcal{E}(\Lambda).$$

Proof. The effective operator has compact resolvent as proved above, so its eigenvectors are in its operator domain, and $\Lambda > 0$ because $e(q) \ge \hbar\omega_*/2$. The exact intertwining identity and residual estimate prove the norm bound. The spectral theorem for H_2 gives $\|(H_2 - \Lambda)u\| \ge \operatorname{dist}(\Lambda, \sigma(H_2))\|u\|$ for every operator-domain vector. Apply it to the normalized Jf. □

The preceding comparison is an actual isometry, an exact compressed Hamiltonian, and an explicit finite-time error estimate for this specific interacting two-coordinate mechanics. At fixed $\hbar, \Gamma, L_1, m^2, E, T$, its stated error tends to zero as $\mathcal{V}_3 \to \infty$, for any family of initial states satisfying the stated energy bound. This is not a volume-uniform mass-gap claim and not a continuum limit. The estimate is conservative; in particular the second-derivative bound deliberately uses the positive quartic energy to control $y^2 R_q$, which is sufficient for the proved $\mathcal{V}_3^{-1/2}$ error without an unproved adiabatic expansion or omission of y^4.

Context and attribution. Moving spectral subspaces, the Berry connection, diagonal Born–Huang energy, and errors in time-dependent Born–Oppenheimer comparisons are established concepts; see G. Panati, H. Spohn, and S. Teufel, *The time-dependent Born–Oppenheimer approximation*, ESAIM: M2AN **41** (2007), 297–314, `https://arxiv.org/abs/0712.4369`, `https://doi.org/10.1051/m2an:2007023`. No new general approximation theorem is claimed here. The spectral bundle, every retained coefficient, and the conservative residual and time estimates used above were proved directly for (12). The bounded-below bridge formula used in the positivity argument is also stated in B. Simon, *A Feynman–Kac Formula for Unbounded Semigroups*, Theorem 1.1, `https://arxiv.org/abs/math-ph/9907022`; its general unbounded-below extension is not needed here.